Decision Trees – Exam Notes

1. What is a Decision Tree?

A Decision Tree is a supervised machine learning model used for both classification and regression tasks. It splits the data into subsets based on feature values using a tree-like structure consisting of a root node, internal nodes, and leaf nodes.

Example: If Income $>$ 50k $\rightarrow$ Yes, else No.

2. ID3 vs C4.5 vs CART

Feature	ID3	C4.5	CART
Measure	Information Gain	Gain Ratio	Gini Index
Split Type	Multiway	Multiway	Binary
Missing Values	No	Yes	Yes
Pruning	No	Yes	Yes
Data Type	Categorical	Both	Both

3. Entropy

Entropy measures the impurity or randomness in a dataset.

$$H(q) = -\sum_{i=1}^{k} p_i \log_2(p_i)$$

Example:

- 10 Yes, 0 No $\rightarrow$ Entropy = 0 (pure node)
- 5 Yes, 5 No $\rightarrow$ Entropy = 1 (maximum impurity)

4. Information Gain

Information Gain measures the reduction in entropy after a split.

$$\text{Gain} = H(\text{parent}) - \sum \frac{N_i}{N} H(\text{child}_i)$$

Example: A split that produces purer child nodes has higher Information Gain.

5. Gain Ratio

Gain Ratio is a normalized version of Information Gain that reduces bias toward attributes with many values.

$$\text{Gain Ratio} = \frac{\text{Information Gain}}{\text{Split Information}}$$

6. Gini Index

Gini Index measures the impurity of a dataset (used in CART).

$$\text{Gini}(q) = \sum_{i=1}^{k} p_i(1 - p_i)$$

Example:

- 50% Yes, 50% No $\rightarrow$ Gini = 0.5
- 100% Yes $\rightarrow$ Gini = 0 (pure)

7. Handling Numeric Attributes

For numeric features, the algorithm finds an optimal threshold v and splits the data as:

- $x < v$ (left child)
- $x \geq v$ (right child)

8. Recursive Partitioning

The process of repeatedly splitting the dataset into smaller subsets until a stopping criterion is met (pure nodes or maximum depth reached).

9. Overfitting

Overfitting occurs when the tree grows too deep and learns noise from the training data, resulting in poor performance on unseen data.

10. Pruning

Pruning reduces overfitting by removing unnecessary branches.

Types of Pruning

Pre-Pruning (Early Stopping):

- Limit maximum depth of the tree
- Set minimum samples required per node
- Set minimum samples per leaf

Post-Pruning:

- Grow the full tree first, then remove branches that do not improve validation performance

11. Detecting Useless Splits

Use statistical tests such as the Chi-Square test to determine whether a split provides statistically significant improvement.

12. Advantages of Decision Trees

- Easy to understand and interpret (white-box model)
- Can handle both numerical and categorical data
- Requires very little data preprocessing
- Fast to train and predict
- Non-parametric (no assumptions about data distribution)

13. Disadvantages of Decision Trees

- Prone to overfitting
- Unstable (small changes in data can result in very different trees)
- Greedy algorithm (does not guarantee globally optimal tree)
- Not suitable for unstructured data (e.g., images, text)
- Can be biased toward features with more levels

14. Classification vs Regression Trees

Aspect	Classification Tree	Regression Tree
Output	Class label	Continuous value
Splitting Measure	Entropy / Gini Index	Squared Error / MSE
Example	Yes/No prediction	House price prediction

15. How to Select the Best Split

Choose the split that maximizes purity (or minimizes impurity):

- Highest Information Gain (ID3)
- Highest Gain Ratio (C4.5)
- Lowest Gini Index (CART)
- Lowest Squared Error (Regression)

Quick Revision

- Entropy = measure of impurity
- Information Gain = reduction in entropy after split
- Gini Index = impurity measure used in CART (faster than entropy)
- Best split = maximum gain or minimum impurity
- Pruning = technique to avoid overfitting

Why is Gini Index Used?

The Gini Index is used in the CART algorithm to measure node impurity and select the best split.

- It calculates the probability of a randomly chosen element being incorrectly classified.
- Lower Gini value $\rightarrow$ higher purity (better split).
- Computationally faster and simpler than Entropy.
- Naturally supports binary splits.

Example:

- Pure node (100% one class) $\rightarrow$ Gini = 0
- Equally mixed classes (50%-50%) $\rightarrow$ Gini = 0.5